

Notes: Xia (JFE, 2014)

Can investor-paid credit rating agencies improve the information quality of issuer-paid rating agencies?

Contents

1	Big picture	3
2	Incremental contribution	3
3	Data and measurement	4
3.1	Rating data and coverage event	4
3.2	Expected default probability (EDP)	4
3.3	Responsiveness of ratings to credit risk	4
3.4	Information content of rating changes	4
3.5	Firm controls	5
4	Models and empirical specifications	5
4.1	Baseline rating-responsiveness specification	5
4.2	Information-content specification	5
4.3	Instrumental-variable strategy	6
4.4	Falsification and matching	6
4.5	Learning versus reputation	6
5	Identification and empirical design	6
5.1	Identifying variation	6
5.2	Threats to identification	7
5.3	Why the evidence is credible	7
6	Tables and figures	7
6.1	Figure 1: EJR's market share and coverage initiation, and Table 1: Rating sample summary statistics	7
6.2	Figure 2: Relation between EDP and credit ratings, and Table 2: Responsiveness of S&P's ratings to credit risk	8
6.3	Table 3: Information content of S&P's rating changes	9
6.4	Table 4: Instrumental variable regressions	9
6.5	Table 5: Exclusion restriction	10
6.6	Table 6: Falsification tests	10
6.7	Table 7: Propensity score matching	11
6.8	Tables 8 and 9: S&P's response to EJR's lower/higher ratings and credit rating standards . .	12
7	Short summary	13
8	Conclusion	13

8.1	Core conclusion	13
8.2	Incremental contribution in one paragraph	14
8.3	Economic intuition	14
8.4	Why the reputation interpretation is persuasive	14
8.5	Policy implications	14
8.6	Limitations and open questions	14
8.7	Takeaway	14

1 Big picture

- **Question.** Does entry by an investor-paid credit rating agency improve or reduce the information quality of incumbent issuer-paid rating agencies?
- **Setting.** The paper studies the Egan-Jones Rating Company (EJR), an investor-paid rating agency, and Standard & Poor's (S&P), an incumbent issuer-paid agency.
- **Core contrast.** Prior work finds that entry by another issuer-paid agency can reduce incumbents' quality because the entrant competes for issuer business. Xia studies a different type of entrant: an investor-paid rater whose ratings can expose weaknesses in issuer-paid ratings rather than compete directly for issuer fees.
- **Main empirical design.** The paper compares S&P rating quality before and after EJR initiates coverage of the same firm. The main sample includes 588 firms for which EJR initiates coverage from 1999Q3 through 2011Q4.
- **Main result 1: responsiveness.** After EJR begins covering a firm, S&P ratings become more responsive to market-based credit risk, measured by expected default probability (EDP) from the Merton/KMV model.
- **Main result 2: information content.** S&P rating downgrades trigger larger negative stock-market reactions after EJR coverage, suggesting that rating changes contain more new information.
- **Main result 3: rating standards.** S&P becomes more conservative and adopts more stringent rating standards following EJR coverage, especially when EJR initially gives a lower rating than S&P.
- **Causal evidence.** The paper uses an instrumental-variable design, exclusion-restriction tests, falsification tests, and propensity-score matching.
- **Mechanism.** The evidence favors a reputation/incentive channel over a pure learning channel. S&P responds strongly when EJR ratings are lower, which is the case in which S&P faces reputational risk from appearing to have overrated the firm.
- **Policy message.** More rating providers do not all have the same effect. An issuer-paid entrant may worsen rating quality by intensifying issuer-friendly competition, while an investor-paid entrant can discipline incumbents by revealing rating-quality problems.

2 Incremental contribution

- **Separates business model from market entry.** The paper's key contribution is to show that the effect of credit-rating competition depends on the entrant's payment model. The effect of EJR is not the same as the effect of Fitch-type issuer-paid competition.
- **Introduces investor-paid entry as a disciplining force.** The paper provides direct evidence that investor-paid coverage can improve incumbent issuer-paid rating quality by increasing reputational pressure.
- **Moves beyond level competition.** Rather than only asking whether an extra rater changes rating levels, the paper studies information quality through responsiveness to credit risk and stock-market information content.
- **Provides a credibility strategy for rating-agency quality.** The use of firm-specific coverage timing, an instrument based on industry-average market capitalization, no-deviation falsification tests, and propensity-score matching is an important empirical contribution.
- **Distinguishes learning from reputation.** The lower-rating versus higher-rating split is a clean conceptual test. Symmetric responses would indicate learning or mimicking; asymmetric responses toward lower EJR ratings indicate reputation and incentives.

- **Adds to the policy debate.** The paper offers evidence that policies encouraging independent/investor-paid rating provision may improve information production, while simply increasing issuer-paid competition may not.

3 Data and measurement

3.1 Rating data and coverage event

- The data combine EJRs ratings with S&P RatingXpress data from July 1999 through 2011.
- The key event is the quarter in which EJR initiates coverage of a firm.
- The main event window is 12 quarters before through 12 quarters after EJR coverage initiation; the figure analysis also tracks a longer post-event window.
- Ratings are converted to a cardinal scale from 1 (AAA) to 21 (C). A higher numerical value means lower credit quality.
- EJR ratings are investor-paid and generally more timely and informative; S&P ratings are issuer-paid and may be influenced by issuer incentives.

Interpretation: The timing of EJR coverage gives a natural before-after design for studying how S&P changes its rating strategy when a more independent rating signal becomes available.

3.2 Expected default probability (EDP)

- EDP is derived from the Merton/KMV framework.
- The model uses market equity value, debt, equity volatility, and the risk-free rate to infer asset value and asset volatility.
- Monthly EDP is aggregated to quarterly frequency.
- EDP is treated as a market-based proxy for fundamental credit risk, less directly subject to issuer-paid incentive distortions.

Interpretation: The paper does not need EDP to be a perfect predictor of default. It needs EDP to be an unbiased market-based benchmark that is not subject to the same issuer-paid rating incentives.

3.3 Responsiveness of ratings to credit risk

- Ratings are more informative if they co-move more strongly with credit risk.
- Because a higher numerical rating value means worse quality, a positive relation between EDP and the rating number indicates responsiveness.
- The key coefficient is the interaction between EJR coverage and EDP.

Interpretation: If S&P improves after EJR coverage, the slope of S&P ratings with respect to EDP should become steeper.

3.4 Information content of rating changes

- The paper studies three-day cumulative abnormal returns (CARs) around S&P rating-change announcements.
- The three-day window includes the day before, the event day, and the day after the rating change.

- Larger negative reactions to downgrades imply that downgrades reveal more new information.
- Upgrade reactions are studied but are less central because positive news is often released voluntarily before upgrades.

Interpretation: Market reactions are a direct test of whether rating changes contain information that investors had not already priced.

3.5 Firm controls

- Controls include leverage, profitability, market-to-book ratio, tangibility, sales, leverage volatility, and profitability volatility.
- The main regressions include firm fixed effects and year dummies when appropriate.
- Standard errors are clustered at the firm level.

Interpretation: Firm fixed effects absorb time-invariant differences across firms, while year effects absorb aggregate rating-market and macro conditions.

4 Models and empirical specifications

4.1 Baseline rating-responsiveness specification

The key baseline model is:

$$Rating_{i,t} = \beta_1 EJRCoverage_{i,t} \times EDP_{i,t} + \beta_2 EDP_{i,t} + \beta_3 EJRCoverage_{i,t} + \beta_4 X_{i,t} + \alpha_i + \alpha_t + \varepsilon_{i,t}. \quad (1)$$

- $Rating_{i,t}$ is S&P rating coded from 1 (AAA) to 21 (C).
- $EJRCoverage_{i,t} = 1$ in the post-EJR coverage period.
- $EDP_{i,t}$ is expected default probability.
- $X_{i,t}$ is the vector of firm characteristics.
- α_i and α_t are firm and year fixed effects.

Interpretation: Since higher EDP implies worse credit risk and higher rating numbers imply worse ratings, $\beta_1 > 0$ means S&P ratings become more responsive to credit risk after EJR coverage.

4.2 Information-content specification

The stock-market reaction test is:

$$CAR_{i,t} = \gamma_1 EJRCoverage_{i,t} + \gamma_2 X_{i,t} + \alpha_t + u_{i,t}. \quad (2)$$

- $CAR_{i,t}$ is the three-day abnormal return around a S&P rating-change announcement.
- The paper focuses on downgrades because they reveal negative credit information.
- A more negative γ_1 for downgrades means that downgrades are more informative after EJR coverage.

Interpretation: If S&P previously delayed or softened negative information, then post-coverage downgrades should surprise investors more and generate larger negative returns.

4.3 Instrumental-variable strategy

The instrument is industry-average market capitalization:

$$IndMeanCap_{-i,t} = \log(\text{average market capitalization of S\&P-rated firms in firm } i\text{'s industry, excluding } i). \quad (3)$$

The first stage predicts EJR coverage timing; the second stage uses instrumented coverage and instrumented interaction terms in the rating-responsiveness and CAR regressions.

Interpretation: The instrument is motivated by EJR’s policy of initiating coverage based on subscriber requests and firm size. Large-cap industries are more likely to attract EJR coverage, but industry-average market capitalization should not directly determine S&P’s rating quality for one particular firm once controls are included.

4.4 Falsification and matching

- **No-deviation falsification.** Firms for which EJR ratings match S&P ratings are treated as cases where EJR enters but does not provide a distinct challenge to S&P quality.
- **Deviation group.** Firms whose EJR ratings differ from S&P ratings are cases where EJR actually challenges S&P’s information.
- **Propensity-score matching.** EJR-rated firms are matched to similar firms not covered by EJR; hypothetical coverage dates are assigned to controls.

Interpretation: If the result is causal, S&P should improve only when EJR provides distinct ratings and only for firms actually covered by EJR, not for matched firms with similar observable characteristics.

4.5 Learning versus reputation

- Learning predicts that S&P should respond symmetrically to EJR’s lower and higher ratings.
- Reputation predicts a stronger response when EJR ratings are lower than S&P ratings because that case exposes potential S&P overrating.
- Overrating is especially costly because downgrades can force investors to sell, lower issuer confidence in ratings, and invite regulatory scrutiny.

Interpretation: The asymmetric response is the core mechanism test. It shows that S&P is not merely copying EJR but reacting to reputational discipline.

5 Identification and empirical design

5.1 Identifying variation

- Variation comes from the timing of EJR coverage initiation across firms.
- The event window compares the same firm before and after EJR coverage.
- Firm fixed effects remove time-invariant firm differences.
- Year effects remove economy-wide changes in rating behavior or credit-market conditions.

Interpretation: The basic identifying assumption is that, absent EJR coverage, S&P’s rating quality for covered firms would not have changed in the same way at the same time.

5.2 Threats to identification

- EJR coverage is not random; investors may request coverage when concerned about a firm's credit quality.
- S&P might also respond to those investor concerns, even without EJR coverage.
- Regulation such as SOX could increase rating quality during the sample period.
- Industry-level changes could affect both EJR coverage and S&P rating behavior.

Interpretation: The paper's design explicitly targets these concerns with IV, falsification, matching, industry-size tests, and pre-2002 subsamples.

5.3 Why the evidence is credible

- The IV results are stronger than OLS, suggesting EJR covers firms whose S&P ratings would otherwise be less informative.
- The instrument does not predict rating quality among non-EJR-rated firms.
- The falsification test shows improvement only when EJR ratings deviate from S&P ratings.
- The propensity-score tests show no similar effect for matched non-covered firms.
- The lower-versus-higher EJR split identifies reputational discipline rather than mechanical learning.

Interpretation: No single test is decisive by itself, but the pattern across tests points to the same mechanism: investor-paid coverage disciplines issuer-paid incumbents.

6 Tables and figures

6.1 Figure 1: EJR's market share and coverage initiation, and Table 1: Rating sample summary statistics

Read:

- **Figure 1 Panel A** shows that EJR's market share grew substantially from 1999 to 2011. Its share by firm count rises from roughly one-quarter of S&P-rated public firms in 1999 to around 60% by 2011.
- The asset-value share is even larger, indicating that EJR tends to cover economically important firms.
- **Figure 1 Panel B** shows that coverage initiation is especially high early in the sample and then remains positive throughout the period.
- **Table 1 Panel A** shows that firms covered by EJR tend to be larger, less levered, less tangible, and more R&D-intensive than the average S&P-rated firm.
- **Table 1 Panel B** shows that EJR ratings are slightly worse on average than S&P ratings on the 1-to-21 scale, consistent with EJR being less issuer-friendly.

Economic message: EJR is not a marginal niche rater. It covers a large share of the market, especially large issuers, and therefore has enough market presence to discipline incumbent rating agencies.

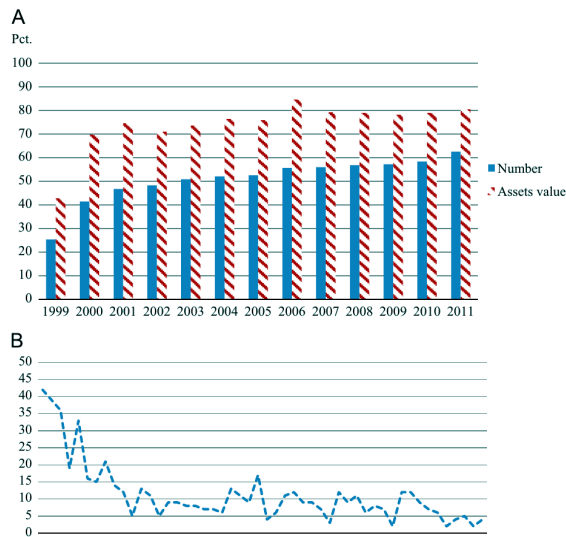


Figure 1: EJR market share and coverage initiation

Table 1
Rating sample summary statistics.
This table presents descriptive statistics for the rating sample between the third fiscal quarter of 1999 and the fourth fiscal quarter of 2011. Panel A reports firm characteristics. Column 1 consists of all U.S. public firms rated by S&P. Column 2 consists of U.S. public firms that are later covered by EJR between 1999 and 2011. Total book assets, Sales, and Market capitalization are in millions of U.S. dollars. Leverage is the ratio of total debt from the balance sheet to total assets; Profitability is the ratio of operating income before depreciation to total assets; M-B ratio is the ratio of the market value of assets to total book value of assets, where the numerator is defined as the sum of market equity and total debt. Tangibility is the ratio of net property, plant, and equipment to total assets; R&D-Sales is the ratio of R&D expense to sales, where R&D expense is replaced by zero if missing. Panel B presents summary statistics for S&P's and EJR's ratings during the 12 quarters following EJR initiating coverage of a firm. ***, **, and * indicate significantly different from zero at the 1%, 5%, and 10% levels, respectively.

Panel A: Firm characteristics								
	All firms rated by S&P (1)			Firms also covered by EJR between 1999 and 2011 (2)			Mean (2)–(1)	Median (2)–(1)
	N	Mean	Median	N	Mean	Median		
Total book assets (\$M)	85,366	12,695	2,619	19,084	15,523	3,719	+++	+++
Sales (\$M)	85,527	1,359	392	19,104	1,729	641	+++	+++
Market capitalization (\$M)	68,439	7,698	1,815	18,393	8,358	2,489	+++	+++
Leverage	85,238	0.383	0.339	19,041	0.348	0.318	+++	+++
Profitability	77,534	0.029	0.028	17,294	0.029	0.028	---	---
M-B ratio	68,327	1.191	0.988	18,157	1.189	0.984	---	---
Tangibility	80,625	0.333	0.270	18,298	0.319	0.252	---	---
R&D-Sales	85,596	0.016	0	19,122	0.021	0	+++	---
Panel B: S&P's and EJR's credit ratings								
N	S&P's ratings (1)		N	EJR's ratings (2)		Mean (2)–(1)		
	Mean	Std. dev.		Mean	Std. dev.			
6,262	11.164 (≈ BB+)	3.517	6,262	11.521	4.093	+++		

Table 1: Rating sample summary statistics

6.2 Figure 2: Relation between EDP and credit ratings, and Table 2: Responsiveness of S&P's ratings to credit risk

Read:

- **Figure 2 Panel A** shows that EJR ratings track the rise and fall in EDP around coverage initiation.
- **Figure 2 Panel B** shows that S&P ratings track EDP weakly before EJR coverage but become more aligned after coverage.
- **Table 2** formalizes this result. The interaction $\text{EJRCoverage} \times \text{EDP}$ is positive and statistically significant in the main specification.
- In column (5), the interaction coefficient is 1.493, implying that a one-standard-deviation increase in EDP is associated with a meaningfully larger downgrade after EJR coverage than before.
- The result survives firm fixed effects, year dummies, and controls for leverage, profitability, market-to-book, tangibility, sales, and volatility measures.

Economic message: S&P ratings become more risk-sensitive after EJR coverage. This is the paper's first major evidence that investor-paid coverage improves incumbent rating quality.

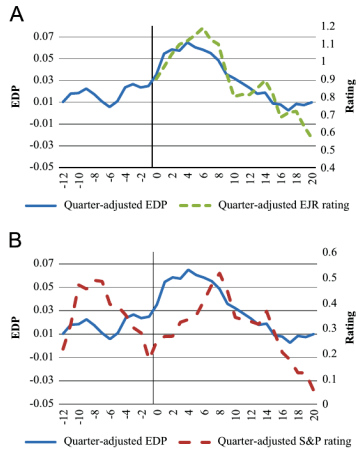


Fig. 2. Relation between the EDP and credit ratings. This figure plots the quarter-adjusted EDP (left y-axis) and credit ratings from S&P and EJR (right y-axis) from 12 quarters prior through 20 quarters following EJR's rating coverage initiation. The sample consists of 588 firms that EJR initiated coverage of between the third fiscal quarter of 1999 and the fourth fiscal quarter of 2011. The x-axis represents the event time (in quarters) where time "0" denotes the quarter when EJR initiates coverage of a firm. Quarter-adjusted EDP (EJR's/S&P's credit ratings) is the difference between the quarterly EDP (EJR's/S&P's credit ratings) and the mean of the quarterly EDP (EJR's/S&P's credit ratings) for all U.S. public firms rated by S&P during the same quarter. The quarterly EDP is measured as the mean of the monthly expected default probability

Figure 2: EDP and ratings

Table 2

Responsiveness of S&P's ratings to credit risk.

This table presents the firm-fixed-effect OLS regression results. The sample consists of firm-quarter observations of 588 firms that EJR initiated coverage of between the third fiscal quarter of 1999 and the fourth fiscal quarter of 2011. The observations are between 12 quarters before and 12 quarters after EJR's coverage initiation. The left-hand-side variable is the cardinal value of S&P's credit ratings from 1 (AAA) to 21 (C). *EJR coverage* is a dummy variable that equals one if a firm-quarter observation is from the post-coverage period (i.e., during the 12 quarters following EJR initiates coverage of the firm), and zero otherwise; *EDP* is the mean of the monthly expected default probability in the past quarter (3 months) where the expected default probability is derived from the Merton/KMV model as in the Appendix. *Leverage* is the ratio of total debt from the balance sheet to total assets; *Profitability* is the ratio of operating income before depreciation to total assets; *M-B ratio* is the ratio of the market value of assets to total book value of assets, where the numerator is defined as the sum of market equity and total debt; *Tangibility* is the ratio of net property, plant, and equipment to total assets; *Sales* is the natural logarithm of sales; all above variables are measured at time $t-1$. *Leverage volatility* is the volatility of leverage during the past four quarters; *Profitability volatility* is the volatility of profitability during the past four quarters. Firm fixed effects are indicator variables for firms; year dummies are indicator variables for fiscal years. Robust standard errors clustered at the firm level are in parentheses. Within-group *R*-squared are reported. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)
EJR coverage*EDP				1.273** (0.500)	1.493*** (0.495)
EDP	2.450*** (0.308)	1.554*** (0.248)	1.299*** (0.255)	0.530* (0.312)	0.671** (0.314)
EJR coverage			0.507*** (0.0738)	0.438*** (0.0739)	-0.0345 (0.0481)
Leverage		3.196*** (0.380)	3.143*** (0.385)	3.091*** (0.384)	3.135*** (0.379)
Profitability		-1.389 (1.674)	-2.555 (1.629)	-2.857* (1.646)	-1.732 (1.679)
M-B ratio		-0.209*** (0.0566)	-0.218*** (0.0556)	-0.220*** (0.0555)	-0.207*** (0.0564)
Tangibility		-0.769 (0.630)	-0.885 (0.638)	-0.912 (0.642)	-0.787 (0.635)
Sales		-0.692*** (0.122)	-0.601*** (0.107)	-0.579*** (0.106)	-0.664*** (0.120)
Leverage volatility		-0.218 (0.501)	-0.435 (0.613)	-0.368 (0.607)	-0.146 (0.595)
Profitability volatility		11.22** (5.634)	10.64* (5.808)	10.52* (5.722)	10.96** (5.501)
Constant	10.75*** (0.0182)	16.00*** (0.869)	13.94*** (0.688)	13.88*** (0.685)	13.38*** (0.677)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year dummies	No	Yes	No	No	Yes
Observations	10,624	9,288	9,288	9,288	9,288
R-squared	0.055	0.252	0.214	0.218	0.259

EJR's coverage initiation. The latter corresponds to a one-notch rating downgrade for approximately one out of every three firms.

percentage) over the three-day event window including one day before the event, the event day (when S&P announces a rating change), and one day following the

Table 2: responsiveness of S&P ratings

6.3 Table 3: Information content of S&P's rating changes

Read:

- Panel A shows that S&P downgrades generate negative stock-market reactions, as expected.
- Downgrade CARs become more negative after EJR coverage: the pre-coverage CAR is about -4.11%, while the post-coverage CAR is about -6.23%.
- The difference is -2.12%, significant at the 10% level in the univariate comparison.
- Upgrade reactions rise from 0.08% to 0.54%, but the difference is not statistically significant.
- Panel B confirms the downgrade result in regressions: the post-coverage effect is negative and statistically significant.

Economic message: S&P downgrades become more informative after EJR coverage. The stronger downgrade result also anticipates the reputation mechanism, because overrating and delayed negative news are especially reputationally costly.

6.4 Table 4: Instrumental variable regressions

Read:

- The first stages show that industry-average market capitalization predicts EJR coverage timing.
- The second-stage rating-responsiveness coefficient on instrumented $EJRCoverage \times EDP$ is positive and significant.
- The second-stage information-content test shows a more negative CAR effect for instrumented EJR coverage.
- The IV estimates are larger than the OLS estimates, suggesting that EJR covers firms whose S&P ratings would otherwise be even less informative.

Table 3
Information content of S&P's rating changes.

This table presents the three-day cumulative abnormal returns (in percentage) surrounding S&P's rating change announcements. The sample consists of S&P's rating change announcements for 588 firms that EJR initiated coverage of between the third fiscal quarter of 1999 and the fourth fiscal quarter of 2011. S&P's rating changes are between 12 quarters before and 12 quarters after EJR's coverage initiation. Panel A presents univariate tests of CARs for downgrades and upgrades separately. Panel B presents the OLS regression results for rating downgrades. The left-hand-side variable is the three-day cumulative stock market abnormal returns surrounding S&P's rating downgrade announcements. EJR coverage is a dummy variable that equals one if an observation is from the post-coverage period (i.e., during the 12 quarters following EJR initiates coverage of the firm), and equals zero otherwise. Change is the number of notches by which S&P changes a firm's rating. Days is the logarithm of the number of days since the previous rating change in the same direction. Lagged S&P ratings is the firm's S&P rating before the rating change. Leverage is the ratio of total debt from the balance sheet to total assets. Profitability is the ratio of operating income before depreciation to total assets. M-B ratio is the ratio of the market value of assets to total book value of assets, where the numerator is defined as the sum of market equity and total debt. Tangibility is the ratio of net property, plant, and equipment to total assets. Sales is the natural logarithm of sales; all firm characteristic variables are measured at time $t-1$. Year dummies are indicator variables for fiscal years. Robust standard errors clustered at the firm level are in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels respectively.

Panel A: Univariate tests—Downgrades and upgrades						
Downgrades				Upgrades		
(1)	(2)	(3)		(4)	(5)	(6)
All obs.	EJR coverage=0	EJR coverage=1	Difference	All obs.	EJR coverage=0	EJR coverage=1
Obs.=658	Obs.=258	Obs.=400	(3)–(2)	Obs.=306	Obs.=128	(6)–(5)
–5.400***	–4.113***	–6.230***	–2.117*	0.348	0.077	0.543
(0.592)	(0.135)	(0.809)	(1.211)	(0.358)	(0.543)	(0.547)
Panel B: Multivariate tests—Downgrades						
	(1)	(2)				
EJR coverage	–2.092*	–2.332*				
	(1.150)	(1.384)				
Change	–3.709***	–3.578***				
	(0.885)	(1.061)				
Days	0.676	–0.197				
	(0.445)	(0.572)				
Lagged S&P rating		–0.363				
		(0.306)				
Leverage		–2.598				
		(3.546)				
Profitability		38.36				
		(31.38)				
M-B ratio		3.002**				
		(1.265)				
Tangibility		1.043				
		(2.955)				
Sales		–0.350				
		(0.054)				
Constant	–2.467	16.21				
	(3.218)	(10.40)				
Year dummies	No	Yes				

Table 4
Instrumental variable regressions.

This table presents the results of the instrumental variables regressions following Cohen, Frazzini, and Malloy (2012). The sample consists of observations of 588 firms that EJR initiated coverage of between the third fiscal quarter of 1999 and the fourth fiscal quarter of 2011. The observations are between 12 quarters before and 12 quarters after EJR's coverage initiation. The first stages are least regressions of the EJR coverage dummy on Ind-Mean-Cap. EJR coverage is a dummy variable that equals one if an observation is from the post-coverage period (i.e., during the 12 quarters following EJR initiates coverage of the firm), and equals zero otherwise. Ind-Mean-Cap is the industry-average market capitalization (measured as the logarithm of market capitalization) of each firm's industry in the concurrent quarter, calculated using only S&P-rated firms not covered by EJR as of 2011. Also included in the first stage are the same control variables as those in the corresponding second stage. Industries are classified by the first-two-digit SIC codes. The firm-quarter with the highest predicted probability of EJR's coverage initiation for each firm is assigned as the instrumented quarter for EJR's coverage initiation. The instrumented EJR coverage equals one if the firm-quarter is after the instrumented quarter for EJR's coverage initiation, and zero otherwise. In the second stage regression in column 2, the left-hand-side variable is the cardinal value of S&P's credit ratings from 1 (AAA) to 21 (C). In the second-stage regressor in column 4, the left-hand-side variable is the three-day cumulative stock market abnormal returns surrounding S&P's rating downgrade announcements. All other variables are defined as in Tables 2 and 3. Robust standard errors clustered at the firm level are in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

	Responsiveness of S&P's ratings to credit risk		Information content of S&P's rating changes	
	(1) First stage	(2) Second stage	(3) First stage	(4) Second stage
EJR coverage×EDP (instrumented)		1.837*** (0.471)		
EDP		0.456 (0.286)		
EJR coverage (instrumented)		0.0656 (0.0659)		–2.212* (1.157)
Ind-Mean-Cap	0.769*** (0.166)	N/A	0.695** (0.313)	N/A
Leverage	1.105 (1.066)	3.109*** (0.382)	0.987 (1.155)	–5.301 (3.469)
Profitability	–26.23*** (5.726)	–2.145 (1.609)	40.48 (25.50)	
M-B ratio	–0.673*** (0.129)	–0.141*** (0.0504)	–0.0777 (1.097)	3.509*** (1.097)
Tangibility	–3.213*** (1.623)	–0.844 (0.643)	–0.127 (2.277)	1.299 (2.941)
Sales	3.523*** (0.572)	–0.642*** (0.121)	2.218*** (0.736)	–0.107 (0.555)
Leverage volatility	–5.413*** (1.399)	–0.0139 (0.629)		
Profitability volatility	6.080 (7.595)	6.546 (5.007)		
Change			0.0007 (0.0545)	–2.833*** (0.899)
Days			–0.427*** (0.106)	0.118 (0.537)
Firm fixed effects	Yes	Yes	Yes	No
Year dummies	No	Yes	No	Yes
Observations	8,606	9,011	495	543
Wald Chi-squared	104.61		31.18	
Prob (Chi-squared)	0.00		0.00	
(Pseudo) R-squared	0.193	0.245	0.173	0.110

Economic message: The IV results address the concern that EJR merely arrives when S&P would have improved anyway. The estimates support a causal effect of EJR coverage on S&P rating quality.

6.5 Table 5: Exclusion restriction

Read:

- Table 5 tests whether industry-average market capitalization directly predicts S&P rating quality among firms not covered by EJR.
- Panel A examines rating responsiveness to credit risk; Panel B examines information content of S&P rating changes.
- Across the full non-EJR sample, firm-size quartiles, matched samples, and top EJR industries, the instrument does not produce the same pattern.
- The coefficients are economically small or statistically insignificant.

Economic message: This is the paper's key exclusion-restriction support. It shows that the instrument is not simply a proxy for industry conditions that mechanically make S&P ratings more informative.

6.6 Table 6: Falsification tests

Read:

- The deviation group consists of firms whose EJR ratings differ from S&P ratings.
- The no-deviation group consists of firms whose EJR ratings are largely identical to S&P ratings.
- In the deviation group, EJR Coverage $\times$ EDP is positive and significant, and downgrade CARs become more negative after EJR coverage.
- In the no-deviation group, neither rating responsiveness nor information content improves.
- This directly tests whether EJR's distinct information, rather than mere coverage timing, drives the effect.

Economic message: S&P reacts only when EJR challenges its rating quality. This is strong evidence against the idea that any investor-requested coverage event would have produced the same S&P response.

Table 5

Exclusion restriction.

This table presents OLS regression results. The sample consists of observations of S&P-rated firms that are not covered by EJR as of fiscal year 2011. In Panel A, the left-hand-side variable is the cardinal value of S&P's credit ratings from 1 (AAA) to 21 (C). In Panel B, the left-hand-side variable is the three-day cumulative stock market abnormal returns surrounding S&P's rating downgrade announcements. In both panels, column 1 consists of the full sample; columns 2-5 consist of subsamples based on the four quartiles of firms' market capitalization, respectively; column 6 consists of the 1-to-3 nearest-neighbor matched sample of firms based on the propensity score matching described in Section 16.1 in the Internet Appendix; column 7 consists of the matched sample firms that are in the top five industries in which EJR rates most firms as of 2011, where industries are classified by the first-two-digit SIC codes. *Ind-Mean-Cap* is the industry-average market capitalization (measured as the logarithm of market capitalization) of each firm's industry in the concurrent quarter, omitting firms that contribute to the average. In Panel A, the control variables are the same as those in column 2 of Table 4; in Panel B the control variables are the same as those in column 4 of Table 4. Robust standard errors clustered at the firm level are in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

	Whole sample	Subsamples of various firm sizes				Matched sample (1-to-3)	Matched sample in the top five EJR-rated industries
	(1)	Smaller than the 25% quartile (2)	Between the 25% and 50% quartile (3)	Between the 50% and 75% quartile (4)	Larger than the 75% quartile (5)	(6)	(7)
Panel A: Responsiveness of S&P's ratings to credit risk							
Ind-Mean-Cap	-0.215	-0.302	-0.440	0.396	-0.251	-0.00873	0.526
	(0.201)	(0.243)	(0.352)	(0.343)	(0.743)	(0.242)	(0.806)
EDP	3.602**	3.739**	4.393	-2.561	3.373	1.607	0.253
	(1.589)	(1.860)	(2.892)	(2.834)	(6.468)	(1.888)	(0.179)
Ind-Mean-Cap	-0.0589	0.0499	0.135	0.0816	0.0341	0.152	-2.316
	(0.0734)	(0.157)	(0.121)	(0.0849)	(0.0941)	(0.113)	(6.439)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	24,214	5,789	5,464	5,562	6,846	28,408	9,273
Within-group R-squared	0.197	0.289	0.213	0.106	0.103	0.176	0.286
Panel B: Information content of S&P's rating changes							
Ind-Mean-Cap	0.0445	-1.101	-0.449	-0.449	0.758	-0.489	-3.008
	(0.847)	(2.057)	(1.600)	(1.293)	(1.505)	(1.087)	(2.866)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	809	210	197	198	198	526	204
Within-group R-squared	0.072	0.174	0.093	0.079	0.153	0.058	0.204

(e.g., credit quality, institutional holdings, and growth) in the top five industries in which EJR rates most firms as

Table 6

Falsification tests.

This table presents falsification test results. The sample consists of observations of 588 firms that EJR initiated coverage of between the third fiscal quarter of 1999 and the fourth fiscal quarter of 2011. The observations are between 12 quarters before and 12 quarters after EJR's coverage initiation. In columns 1 and 2, the left-hand-side variable is the cardinal value of S&P's credit ratings from 1 (AAA) to 21 (C). In columns 3 and 4, the left-hand-side variable is the three-day cumulative stock market abnormal returns surrounding S&P's rating downgrade announcements. Columns 1 and 3 consist of firms whose initial EJR ratings are different from S&P's concurrent ratings; columns 2 and 4 consist of firms whose initial EJR ratings are identical to S&P's concurrent ratings. In columns 1 and 2, all control variables are the same as those in column 5 of Table 2. In columns 3 and 4, all control variables are the same as those in column 2 of Table 3 Panel B. Robust standard errors clustered at the firm level are in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

	Responsiveness of S&P's ratings to credit risk		Information content of S&P's rating changes	
	(1) Deviation	(2) No-deviation	(3) Deviation	(4) No-deviation
EJR coverage	1.527**	0.442		
	(0.602)	(0.684)		
EDP	0.673*	1.141**		
	(0.369)	(0.467)		
EJR coverage	-0.0329	-0.0398	-3.468**	0.648
	(0.0643)	(0.0558)	(1.755)	(4.385)
Control variables	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	No	No
Year dummies	Yes	Yes	Yes	Yes
Observations	6,339	2,949	366	98
(Within-group) R-squared	0.315	0.130	0.119	0.240

"incentive/reputation" channel). More specifically, as EJR initiates coverage, "incentive" and "learning" channels might generate different

6.7 Table 7: Propensity score matching

Read:

- Each EJR-rated firm is matched to one, three, or five similar non-EJR-rated firms.
- The matched controls receive a hypothetical coverage date corresponding to the treated firm's EJR event time.
- The actual EJR sample reproduces the main results in column (1).
- The matched-control samples show no comparable improvement in S&P rating responsiveness or information content.
- The finding holds across 1-to-1, 1-to-3, and 1-to-5 nearest-neighbor matches.

Economic message: The matching test shows that the effect is not driven by observable firm characteristics or event-time patterns. Actual investor-paid coverage matters.

Table 7

Propensity score matching.

This table presents the results of the nearest-neighbor propensity score matching where each EJR-rated firm is matched with one, three, or five firm(s) that are not rated by EJR as of the end of 2011, respectively. The EJR-rated sample consists of observations of 588 firms that EJR initiated coverage of between the third fiscal quarter of 1999 and the fourth fiscal quarter of 2011, and these observations are between 12 quarters before and 12 quarters after EJR's coverage initiation. The detailed description of the matching procedure is in the Internet Appendix, [Section IA.3](#). Firms are matched based on their pre-treated (one-quarter prior to EJR's coverage) firm characteristics and the mean of the one-quarter percentage change of firm characteristics during the four quarters prior to EJR's coverage. The characteristics include EDPs, leverage, profitability, size, institutional investors' stock ownership, market-to-book ratios, and tangibility. The firm-quarter from the non-EJR-rated (control) group that is matched with EJR-rated (treated) firm is applied as the hypothetical event time for the firm. *Hypothetical EJR coverage* equals one if the observation is after the hypothetical event time, and zero otherwise. In Panel A, the left-hand-side variable is the cardinal value of S&P's credit ratings from 1 (AAA) to 21 (C). In Panel B, the left-hand-side variable is the three-day cumulative stock market abnormal returns surrounding S&P's rating downgrade announcements. Column 1 of Panels A and B repeats column 5 in [Table 2](#) and column 2 in [Table 3](#) Panel B for comparison purposes. Columns 2–4 consist of the 1-to-1, 1-to-3, and 1-to-5 matched sample, respectively. In Panel A, all control variables are the same as those in column 5 of [Table 2](#). In Panel B, all control variables are the same as those in column 2 of [Table 3](#) Panel B. Robust standard errors clustered at the firm level are in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2) 1-to-1	(3) 1-to-3	(4) 1-to-5
<i>Panel A: Responsiveness of S&P's ratings to credit risk</i>				
(Hypothetical) EJR coverage*EDP	1.493*** (0.495)	0.202 (0.171)	0.183 (0.164)	0.239 (0.205)
EDP	0.671** (0.314)	0.406** (0.198)	0.803*** (0.178)	1.109*** (0.270)
(Hypothetical) EJR coverage	−0.0345 (0.0481)	−0.0801 (0.0758)	−0.0435 (0.0412)	−0.0426 (0.0334)
Control variables	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes
Observations	9,288	5,432	15,794	26,126
Within-group R-squared	0.259	0.245	0.159	0.179
<i>Panel B: Information content of S&P's rating changes</i>				
(Hypothetical) EJR coverage	−2.332* (1.384)	−2.152 (2.142)	0.739 (1.789)	−0.111 (1.411)
Control variables	Yes	Yes	Yes	Yes

Table 7: Propensity score matching

6.8 Tables 8 and 9: S&P's response to EJR's lower/higher ratings and credit rating standards

Read:

- **Table 8** compares cases where EJR ratings are lower than S&P ratings with cases where they are higher.
- When EJR ratings are lower, S&P becomes more responsive to EDP and downgrades carry more information.
- When EJR ratings are higher, the corresponding effects are absent.
- This asymmetry rejects a simple learning story because learning would predict responses to both positive and negative EJR information.
- **Table 9** shows that S&P rating standards become more stringent after EJR coverage, especially when EJR ratings are lower than S&P ratings.
- In the lower-rating subsample, the coefficient implies roughly a one-notch downward adjustment in S&P ratings for one out of every five firms.

Economic message: These tables are the main mechanism evidence. S&P responds most strongly when EJR exposes possible overrating, which is where reputational costs are highest.

	Responsiveness of S&P's ratings to credit risk		Information content of S&P's rating changes	
	(1) EJR's ratings lower than S&P's	(2) EJR's ratings higher than S&P's	(3) EJR's ratings lower than S&P's	(4) EJR's ratings higher than S&P's
EJR coverage*EDP	1.503** (0.726)	0.707 (0.843)		
EDP	0.781 (0.484)	0.851 (0.619)		
EJR coverage	0.0388 (0.104)	-0.0917 (0.0740)	-3.468** (1.755)	-0.118 (1.231)
Control variables	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	No	No
Year dummies	Yes	Yes	Yes	Yes
Observations	3,600	2,655	366	103
(Within-group) R-squared	0.392	0.296	0.119	0.530

Table 8: S&P response to lower/higher EJR ratings

years. Robust standard errors clustered at the firm level are in parentheses. Within-group *R*-squared are reported. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2) EJR's ratings lower than S&P's	(3) EJR's ratings higher than S&P's
EJR coverage	0.0815* (0.0483)	0.239*** (0.0916)	-0.0744 (0.0799)
Leverage	3.334*** (0.385)	3.072*** (0.677)	3.748*** (0.697)
Profitability	-2.465 (1.764)	-6.345* (3.222)	3.344* (1.964)
M-B ratio	-0.272*** (0.0566)	-0.247** (0.0984)	-0.420*** (0.0906)
Tangibility	-0.569 (0.642)	-0.261 (0.856)	-1.915 (1.187)
Sales	-0.694*** (0.127)	-0.488** (0.211)	-0.672*** (0.148)
Leverage volatility	-0.0256 (0.601)	0.0658 (1.114)	0.260 (1.089)
Profitability volatility	12.78** (5.954)	19.91* (10.12)	7.241 (5.596)
Constant	13.56*** (0.709)	11.78*** (1.184)	14.73*** (0.949)
Firm fixed effects	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes

Table 9: credit rating standards and EJR coverage

7 Short summary

- The paper studies whether investor-paid rating agencies improve the information quality of issuer-paid incumbents.
- The key event is EJR initiating coverage of a firm.
- S&P ratings become more responsive to credit risk after EJR coverage.
- S&P downgrades generate larger negative stock returns after coverage, implying greater information content.
- S&P adopts more stringent rating standards after coverage.
- The causal evidence includes IV, exclusion-restriction tests, falsification tests, and propensity-score matching.
- Effects occur when EJR provides distinct ratings, not when EJR ratings merely replicate S&P ratings.
- Effects are stronger when EJR gives a lower rating than S&P, consistent with reputational pressure rather than symmetric learning.
- The paper's main policy implication is that the payment model of new rating providers matters.

8 Conclusion

8.1 Core conclusion

Xia shows that investor-paid rating coverage can discipline issuer-paid incumbent rating agencies. After EJR initiates coverage, S&P ratings better track market-based credit risk, S&P downgrades contain more information, and S&P adopts more conservative rating standards. The result is not simply that more raters improve markets; it is that investor-paid raters can change incumbent incentives by revealing the quality of incumbent ratings.

8.2 Incremental contribution in one paragraph

The paper’s incremental contribution is to distinguish between competition from issuer-paid entrants and discipline from investor-paid entrants. Prior evidence on issuer-paid competition suggested that new raters can reduce incumbent quality by intensifying issuer-friendly rating competition. Xia adds that an investor-paid rater can have the opposite effect because it does not compete for issuer fees. Instead, its more informative ratings can expose inflated or stale issuer-paid ratings and raise reputational costs for incumbents. This payment-model distinction is the paper’s central conceptual contribution.

8.3 Economic intuition

Issuer-paid rating agencies face a conflict: issuers value favorable ratings, while investors value accurate ratings. An investor-paid rater has the opposite client base and can reveal when issuer-paid ratings fail to reflect credit risk. Once EJR covers a firm, S&P faces a higher probability that its rating errors will become visible to investors and regulators. This creates an incentive to make ratings more responsive and rating changes more informative.

8.4 Why the reputation interpretation is persuasive

The strongest evidence is the asymmetry between lower and higher EJR ratings. If S&P were simply learning from EJR, S&P should respond to both favorable and unfavorable EJR ratings. Instead, S&P responds mainly when EJR is more pessimistic. This is the setting in which S&P appears to have overrated the firm, and overrating carries higher reputational and regulatory costs than underrating. The asymmetry therefore points to incentive discipline rather than mechanical information copying.

8.5 Policy implications

The paper implies that policies encouraging additional rating providers should distinguish between business models. An issuer-paid entrant may worsen ratings if it competes for issuer business. An investor-paid or independent rater can improve quality if it creates outside scrutiny of incumbent ratings. The implication is not simply “more competition,” but “the right kind of competition.”

8.6 Limitations and open questions

The design is strong but not a randomized experiment. EJR coverage is requested by subscribers, and coverage timing may reflect investor concerns. The paper addresses this using multiple empirical strategies, but some unobserved demand for information may remain. The paper also focuses primarily on corporate issuer ratings and on S&P; effects could differ for structured finance, sovereigns, or other rating agencies. Finally, the welfare implications depend on how investors use multiple ratings and whether investor-paid models can scale without creating their own conflicts.

8.7 Takeaway

The paper reframes the debate about rating-agency competition. The central issue is not the number of raters alone but the incentives behind each rater’s business model. Investor-paid rating agencies can improve market information by disciplining issuer-paid incumbents, especially when they reveal negative information that incumbent agencies have incentives to suppress or soften.